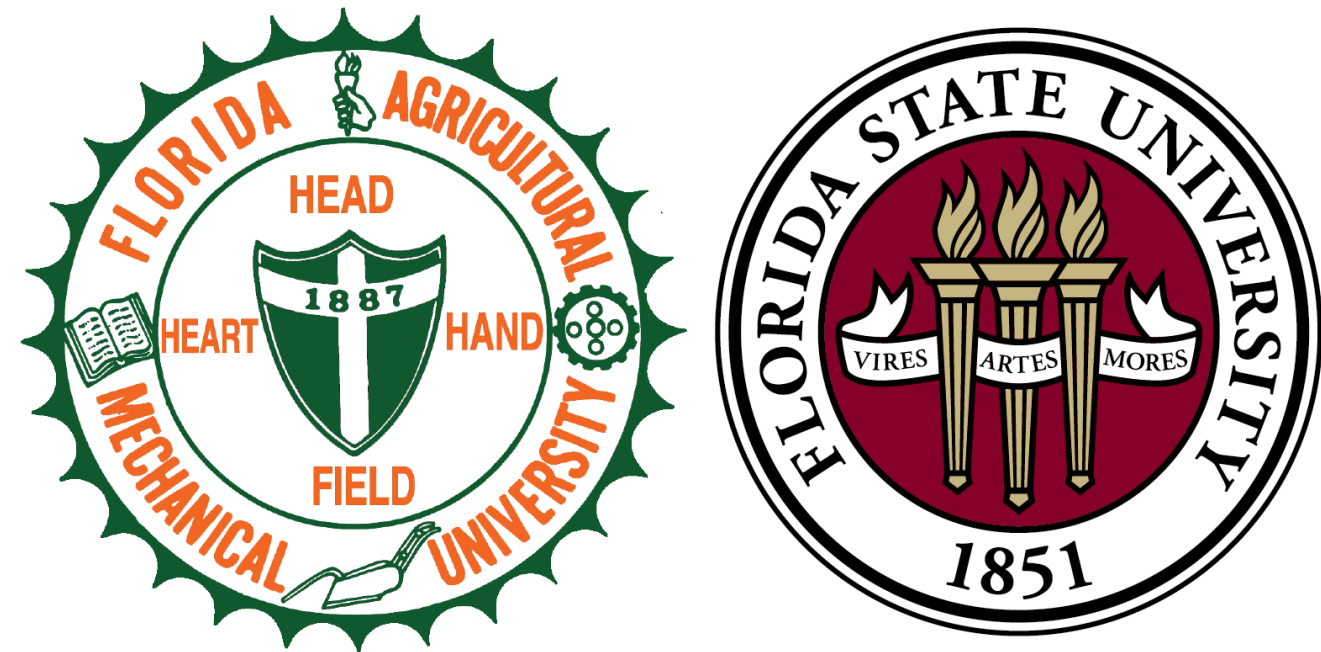


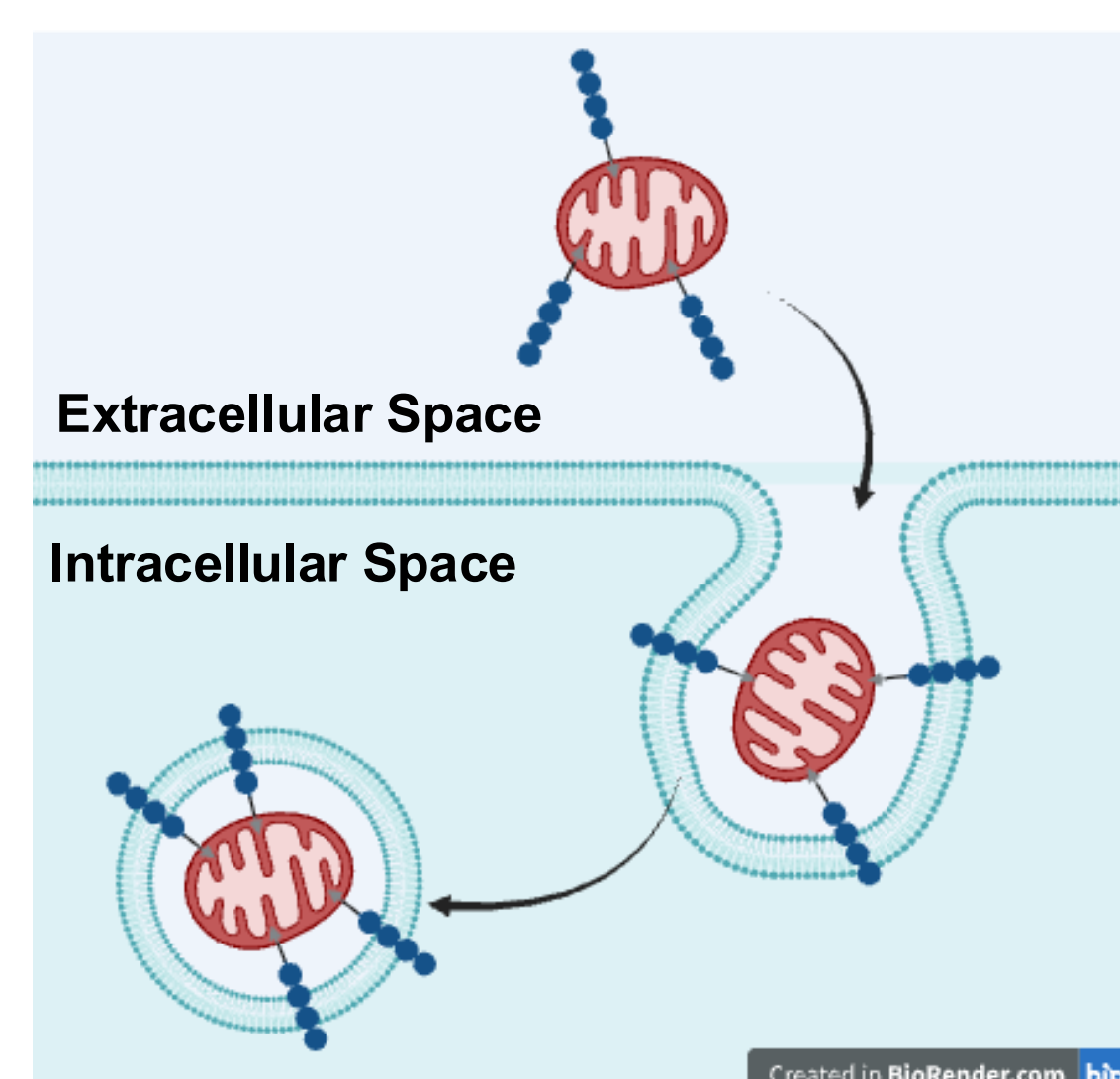


Quantitative Assessment of Mitochondrial Uptake across Cell Populations

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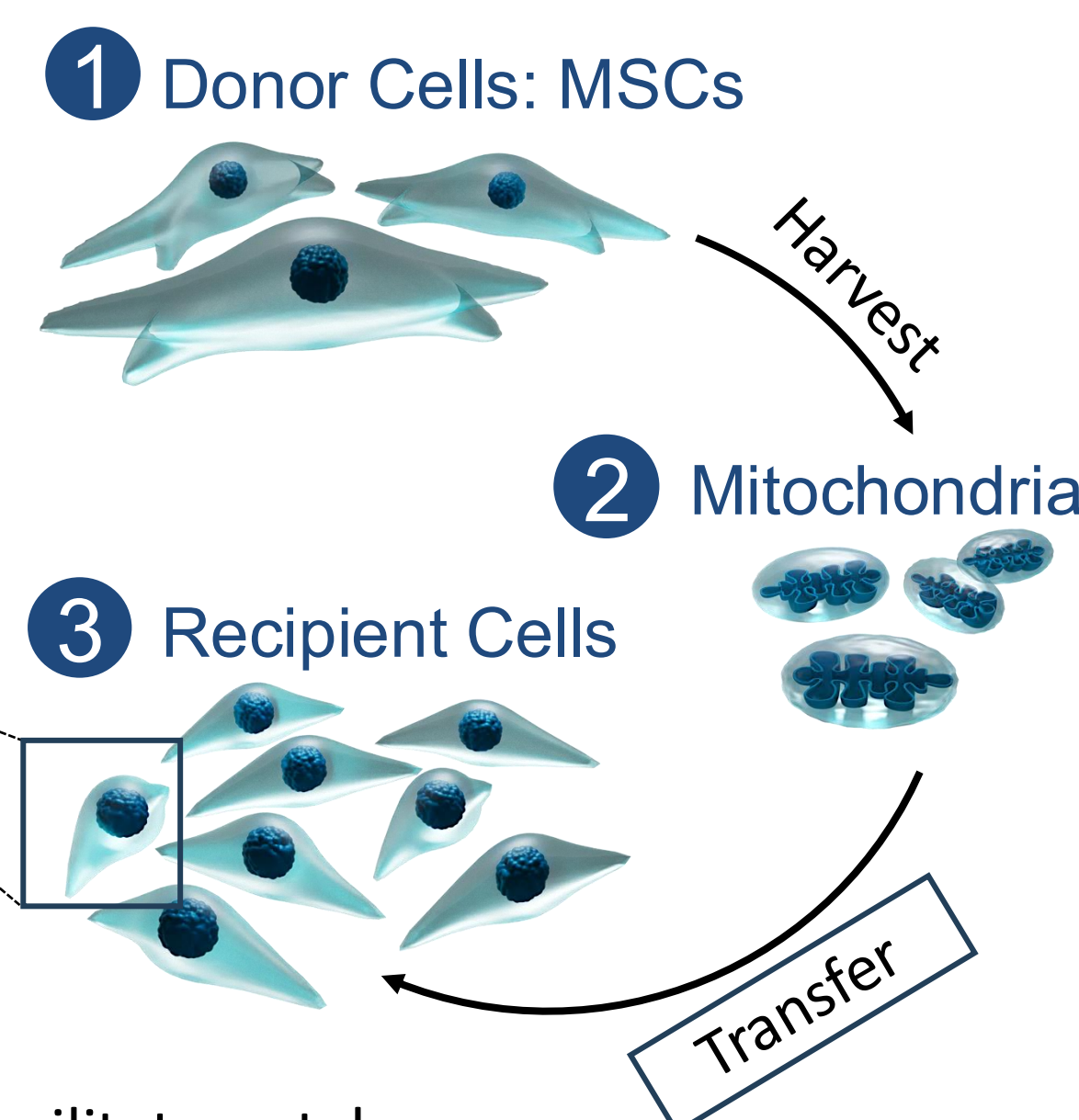
Introduction



- MSCs have been shown to deliver mitochondria to metabolically impaired cells¹.

- Mitochondria can be delivered to cells to increase their bioenergetic state and facilitate tissue regeneration¹.

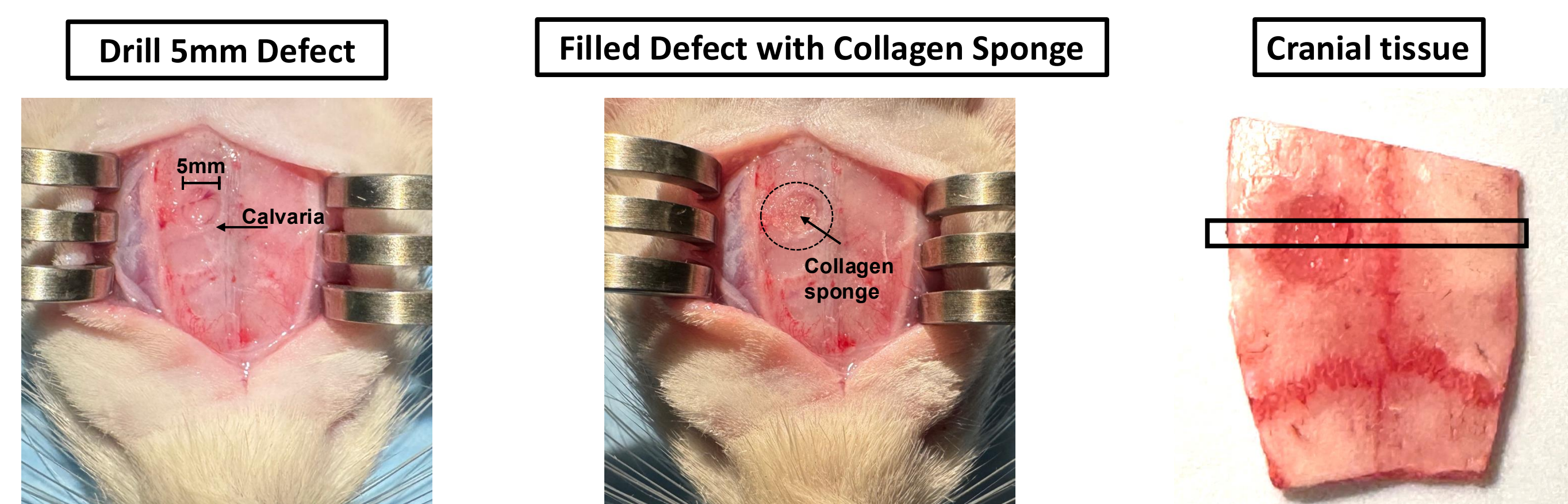
- Mitochondria decorated with a polymer to facilitate uptake.



Quantification of mitochondrial uptake *in vivo* to validate cellular internalization at the injury site.

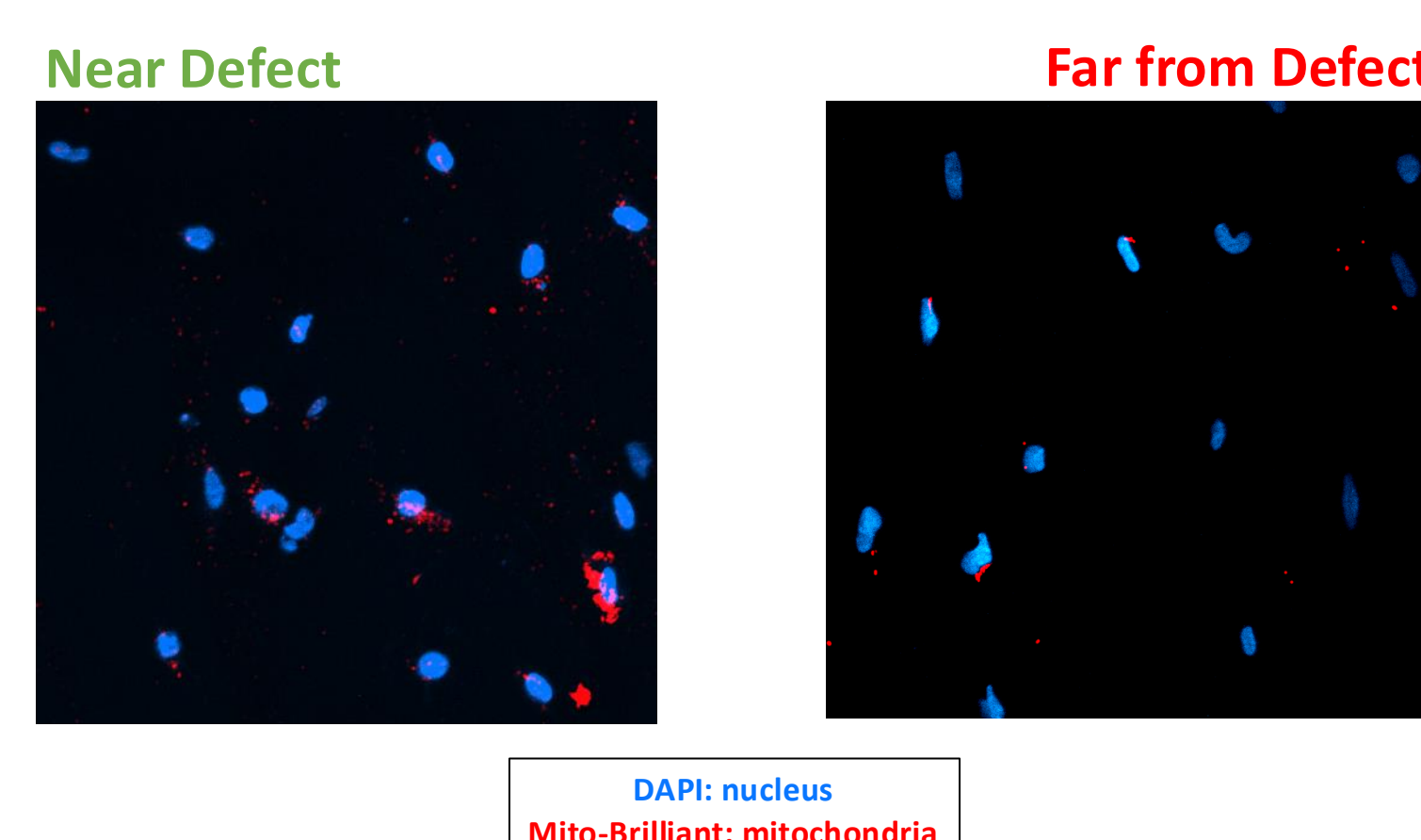
Hypothesis: Mitochondria transplantation in a critical-size calvaria defect is expected to result in greater internalization by tissue-resident cells located near the defect site compared to those farther away.

Mitochondria Transfer in Calvaria Defect rat Model

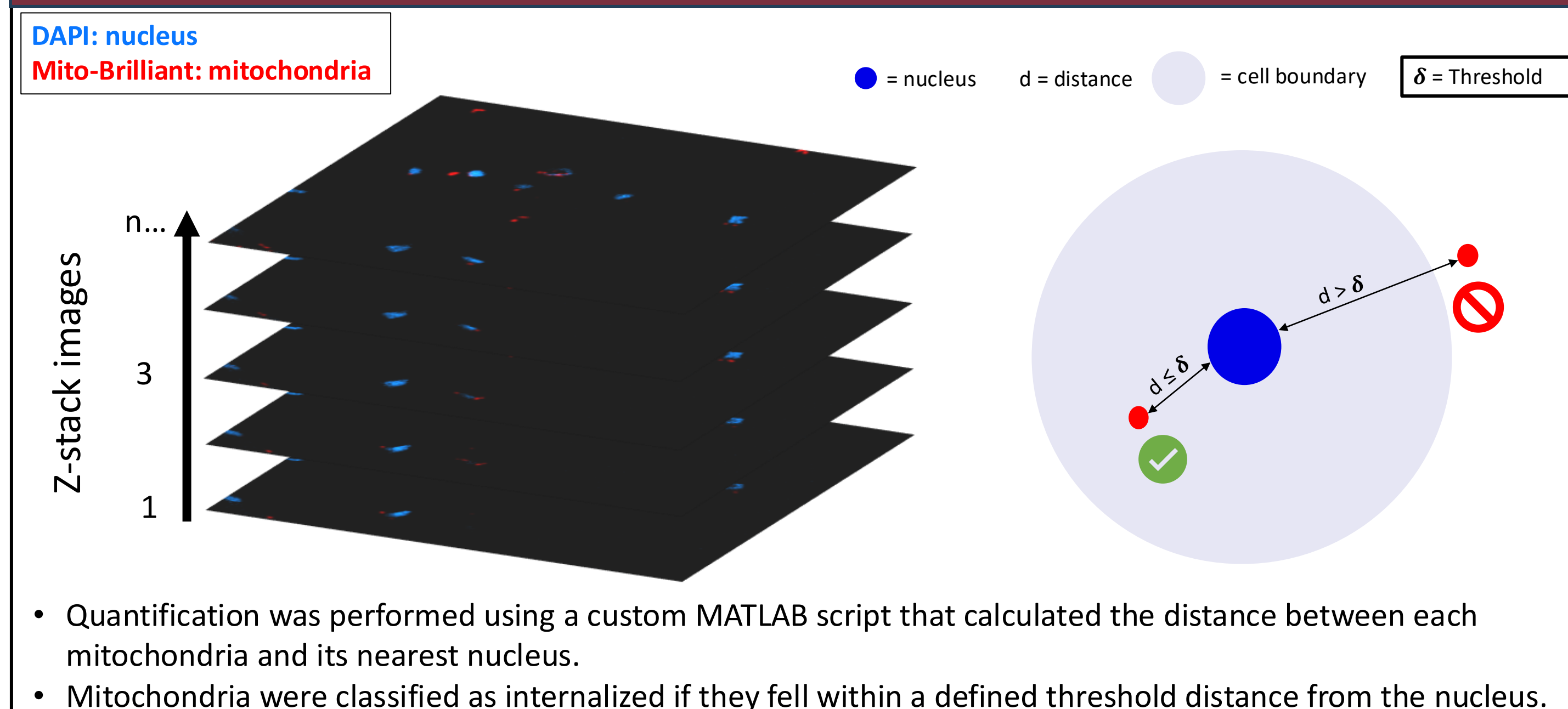


- The rat critical size calvaria defect, a cranial bone injury that does not spontaneously heal, provides a platform to test candidate therapies².

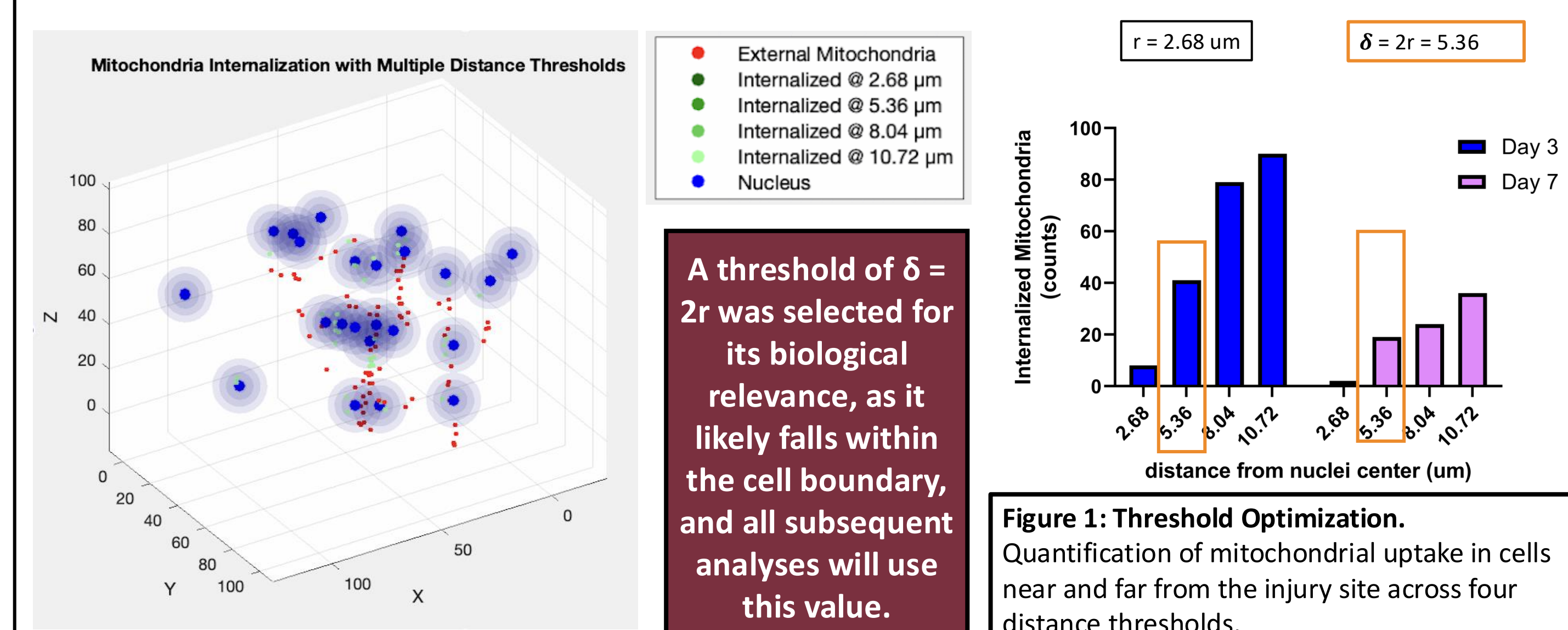
Mitochondria-soaked collagen sponge implanted in calvaria defect; uptake assessed at Day 3 and 7 in near vs. far regions to evaluate spatial delivery and internalization.



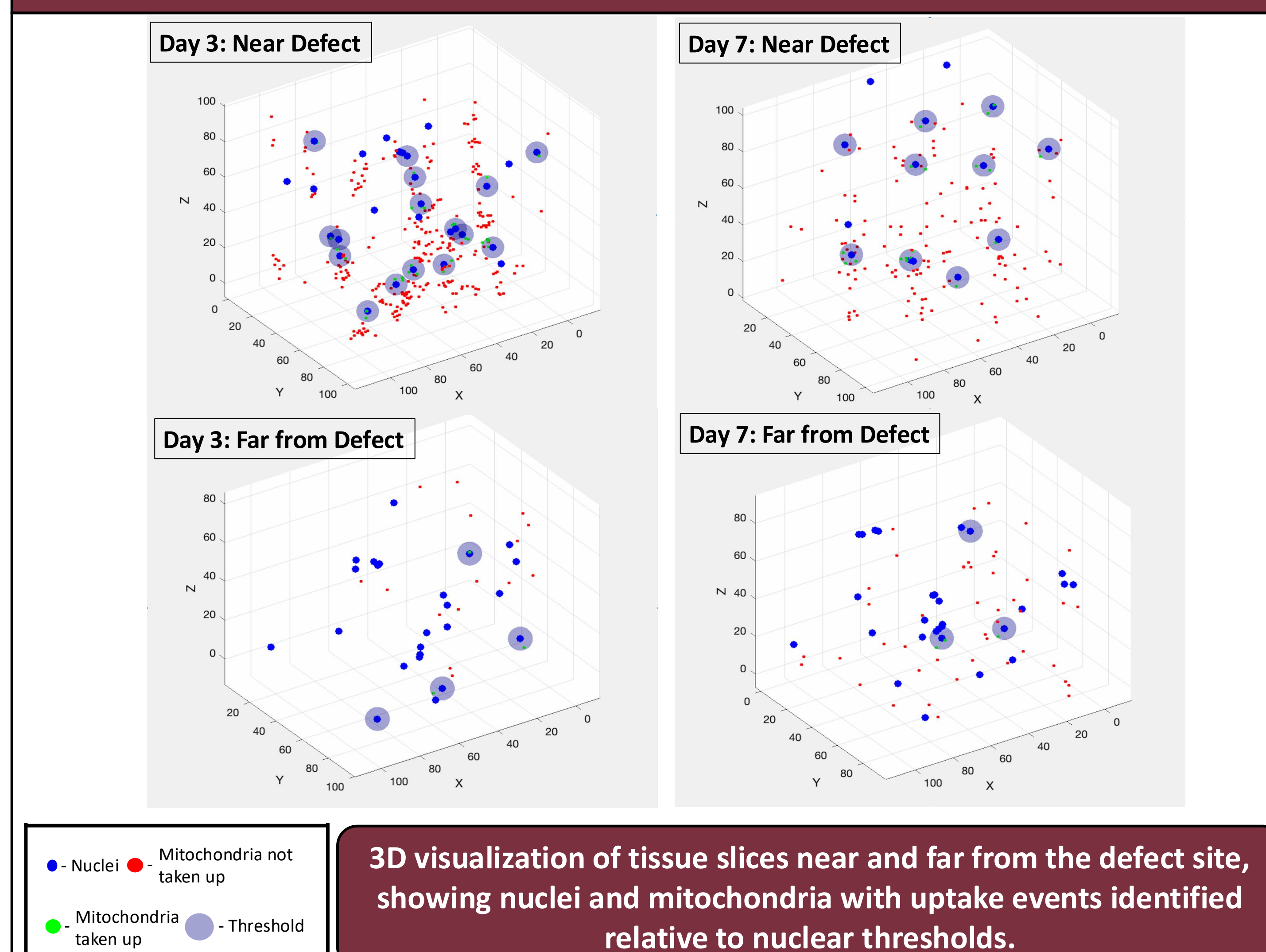
Quantification Method



Threshold Optimization



Visualization of Mitochondria Uptake



Localized Mitochondria Uptake at Defect Region

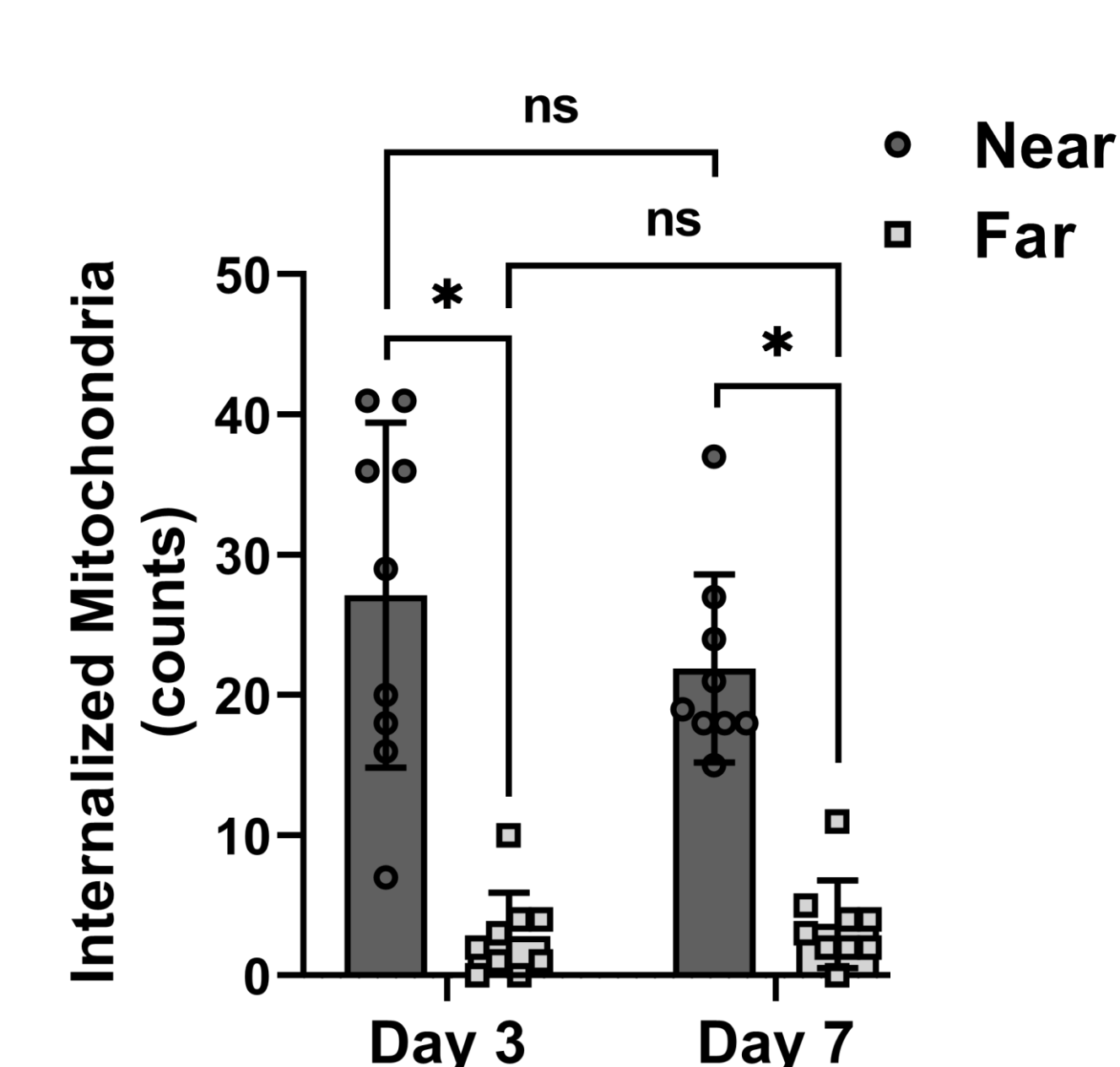


Figure 2: Internalized Mitochondria Near and Far from Defect. Quantification of internalized mitochondria per field of view near vs. far regions at Day 3 and Day 7 (n=9). **Statistics:** * $p < 0.05$, ns = not significant, as determined by either a two-way ANOVA with Tukey's *post hoc* test.

Mitochondrial uptake was greater near the injury site than in distant regions, indicating effective localized delivery.

Substantial Portion of Cells Took Up Mitochondria

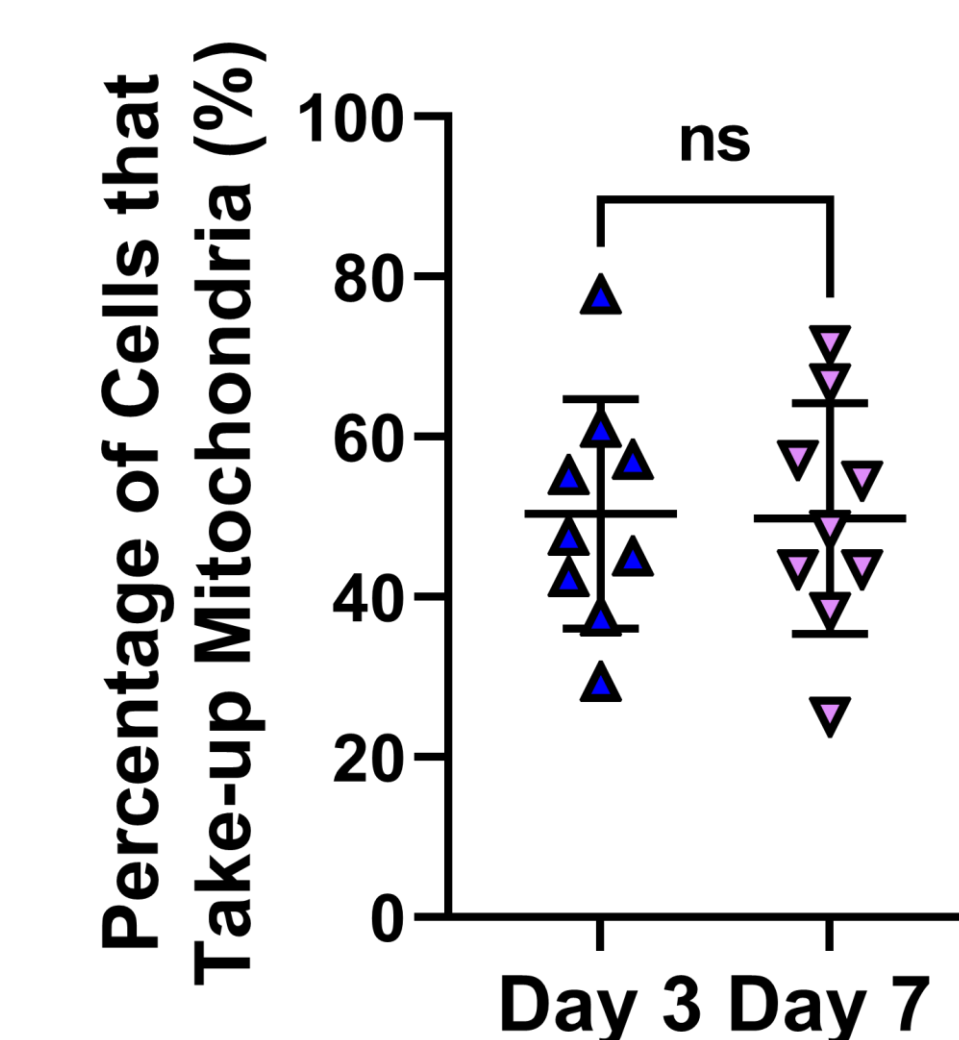
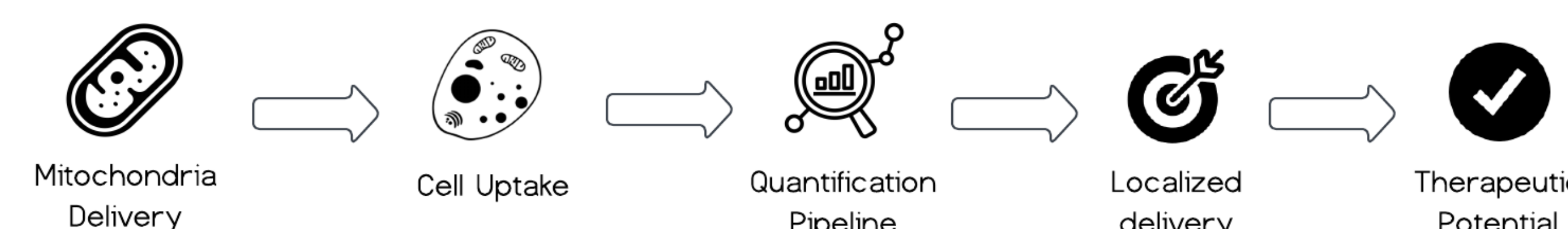


Figure 3: Cellular Uptake. Percentage of cells that internalized mitochondria at Day 3 and Day 7 near the injury site (n = 9). **Statistics:** ns = not significant, as determined by a Mann-Whitney test.

On average, ~50% of cells consistently internalized mitochondria, suggesting broad cellular targeting at the defect site that may support faster tissue repair.

Conclusion



- ✓ Proof of concept achieved.
- ✓ Developed a quantitative pipeline to assess mitochondrial internalization by surrounding cells.
- ✓ Demonstrates proof-of-concept for mitochondrial internalization near defect site, supporting therapeutic potential for tissue repair

ACKNOWLEDGEMENTS

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